STAT2004 - Solutions Week 12

Exercises 1 and 2 were covered in the tutorial.

Exercise 1 (Grades and majors[1])

(a) In words, the null hypothesis is that the statistics grade of the students is independent of, or unrelated to, their majors. The alternative hypothesis is that they are related or associated.

In symbols, this can be written as

$$H_0 : p_{ij} = p_{i\bullet} \times p_{\bullet j} \text{ for } i = 1, 2, 3, 4 \text{ and } j = 1, 2, 3 \ ,$$

where $p_{i\bullet}$ is the overall proportion of students with grade i and $p_{\bullet j}$ is the overall proportion of students in major j. The alternative hypothesis is H_1 : not all joint probabilities p_{ij} factorise into the product of their marginals.

(b) Running the `R` code provided gives us the output:

```
Pearson's Chi-squared test

data:  table1
X-squared = 12.183, df = 6, p-value = 0.058

Warning message:
In chisq.test(table1) : Chi-squared approximation may be incorrect
```

According to this naive analysis, there is moderate-to-marginal evidence that student grades are related to their majors at the 5% level. `R` gave us a warning because some of the expected counts are below 5:

```
chisq.test(table1)$expected
          [,1]      [,2]      [,3]
[1,] 12.203390 11.898305 11.898305
[2,] 16.271186 15.864407 15.864407
[3,]  8.813559  8.593220  8.593220
[4,]  2.711864  2.644068  2.644068
```

(c) In this particular scenario, Fisher's exact test invokes the following logical argument: if grades are indeed unrelated to major, then 36 students would have gotten A's anyway, 48 would have gotten B's anyway, 26 would have gotten C's anyway, and 8 would have gotten D–F anyway, *regardless of their chosen majors*. Similarly, if choice of major is unrelated to statistics grades (or statistical ability), then 40 students would have chosen Psychology anyway, 39 would have chosen Biology anyway, and 39 would have chosen Other anyway, *regardless of their statistical ability*.

Thus, under the null of no association between grades and major, the *margins* of the table would have been fixed but the counts within the table could have been *any* configuration that sums up to the fixed margins.

(d) We can carry out Fisher's exact test on these data using the following command, boosting the number of simulated tables to 100,000, say, which is easily handled on my 2021 laptop:

```
chisq.test(table1, simulate.p.value = T, B=100000)
Pearson's Chi-squared test with simulated p-value (based on 1e+05 replicates)

data:  table1
X-squared = 12.183, df = NA, p-value = 0.05623
```

If there is no association between student grades and their majors then such counts (or configurations that are even more unusual) would have occurred 5.6% of the time by chance alone. There is, at best, only marginal evidence of an association between student grades and their majors.

[1] Adapted from Problem 13.8 of *Mathematical Statistics & Data Analysis* (Rice, 1995)

Exercise 2 (Weight gain and protein diet[2])

(a) The one-way ANOVA model for this scenario is

$$Y_{ji} = \mu_j + \epsilon_{ji} \ , \quad \epsilon_{ji} \overset{\text{iid}}{\sim} N(0, \sigma^2) \ \text{ for } j = 1, 2, 3, \ i = 1, 2, \ldots, 5 \ .$$

(b) $H_0 : \mu_1 = \mu_2 = \mu_3$ versus H_1 : not all group means are the same.

(c) (i)

$$\begin{aligned} \overline{Y}_{\text{Low}\bullet} &= (3.89 + 3.87 + 3.26 + 2.70 + 3.82)/5 \qquad = 3.51\, g \\ \overline{Y}_{\text{Med}\bullet} &= (8.54 + 9.32 + 8.76 + 9.30 + 10.45)/5 \qquad = 9.27\, g \\ \overline{Y}_{\text{High}\bullet} &= (20.39 + 24.22 + 30.91 + 22.78 + 26.33)/5 = 24.93\, g \end{aligned}$$

(ii) $\overline{Y}_{\bullet\bullet} = (3.89 + 3.87 + \ldots + 26.33)/15 = 12.57 g$.

(d) (i) the between-groups sum-of-squares is

$$\begin{aligned} \text{SS}_{\text{diet}} &= 5 \sum_j (\overline{Y}_{j\bullet} - \overline{Y}_{\bullet\bullet})^2 \\ &= 5 \times \left[(3.51 - 12.57)^2 + (9.27 - 12.57)^2 + (24.93 - 12.57)^2 \right] \\ &= 1228.27\, g^2 \end{aligned}$$

(ii) the within-groups residual sum-of-squares is

$$\begin{aligned} \text{SS}_{\text{residual}} &= \sum_{j,i} (Y_{ji} - \overline{Y}_{j\bullet})^2 \\ &= (3.89 - 3.51)^2 + (3.87 - 3.51)^2 + \cdots + (3.82 - 3.51)^2 \\ &\quad + (8.54 - 9.27)^2 + (9.32 - 9.27)^2 + \cdots + (10.45 - 9.27)^2 \\ &\quad + (20.39 - 24.93)^2 + (24.22 - 24.93)^2 + \cdots + (26.33 - 24.93)^2 \\ &= 66.74\, g^2 \end{aligned}$$

(iii) the sum of the group and residual sums-of-squares

$$\text{SS}_{\text{diet}} + \text{SS}_{\text{residual}} = 1228.27 + 66.74 = 1295.01\, g^2$$

(iii) the total sum-of-squares is

$$\begin{aligned} \text{SS}_{\text{total}} &= \sum_{j,i} (Y_{ji} - \overline{Y}_{\bullet\bullet})^2 \\ &= 3.89 - 12.57)^2 + (3.87 - 12.57)^2 + \cdots + (26.33 - 12.57)^2 \\ &= 1295.01\ , \quad \text{the same as in part (iii) above.} \end{aligned}$$

(e) There are 2 gaps between 3 diets so $df_{\text{diet}} = 2$. There are 4 gaps between 5 observations within each group, so there are 4 residual degrees-of-freedom within each group, given $df_{\text{residual}} = 4 \times 3 = 12$. The total degrees-of-freedom is therefore $df_{\text{total}} = 2 + 12 = 14$ – overall, there are 14 gaps between 15 total observations.

(f) Using your calculations from parts (d)–(e), complete the following ANOVA table:

Source	df	SS	MS	F	P(F)
Diet	2	1228.27	614.14	110.43	1.873×10^{-8}
Residual	12	66.74	5.56		
Total	14	1295.01			

We can conclude that there is very strong evidence that the mean weight gain in mice varied by protein level in diets.

(g) Using the `R` code provided gives us the following output:

```
Analysis of Variance Table

Response: weight.gain
          Df  Sum Sq Mean Sq F value    Pr(>F)
diet       2 1228.27  614.14  110.43 1.873e-08 ***
Residuals 12   66.74    5.56
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

[2] Adapted from Exercise 11.21 of *Statistical Inference* (Casella & Berger, 2002)

Exercise 3 (Two-group ANOVA versus two-sample t test)

Consider the following one-way ANOVA model,

$$Y_{ji} = \mu_j + \epsilon_{ji}\ , \quad j = 1, 2,\ i = 1, 2, \ldots, r_j\ ,$$

where $\epsilon_{ji} \overset{\text{iid}}{\sim} N(0, \sigma^2)$ across both indices j and i. Note that the number of replications are not necessarily equal in the two groups.

(a) For the special case of two groups, we have

$$\begin{aligned} \mathrm{SS}_{\mathrm{group}} &= r_1(\overline{Y}_{1\bullet} - \overline{Y}_{\bullet\bullet})^2 + r_2(\overline{Y}_{2\bullet} - \overline{Y}_{\bullet\bullet})^2 \\ \mathrm{SS}_{\mathrm{resid}} &= \sum_{i=1}^{r_1}(Y_{1i} - \overline{Y}_{1\bullet})^2 + \sum_{i=1}^{r_2}(Y_{2i} - \overline{Y}_{2\bullet})^2 \end{aligned}$$

(b) There is one gap between two groups and so $df_{\mathrm{group}} = 1$ for the between-group sum-of-squares. Within each of the two groups, there are $r_1 - 1$ and $r_2 - 1$ gaps respectively, and so $df_{\mathrm{residual}} = r_1 + r_2 - 2$ for the residual sum-of-squares.

(c) Note that the overall mean is equal to the weighted average of the two sample means,

$$\overline{Y}_{\bullet\bullet} = \frac{r_1\overline{Y}_{1\bullet} + r_2\overline{Y}_{2\bullet}}{r_1 + r_2} \ ,$$

so that the between-group sum-of-squares can be written as

$$\begin{aligned} \mathrm{SS}_{\mathrm{group}} &= r_1(\overline{Y}_{1\bullet} - \overline{Y}_{\bullet\bullet})^2 + r_2(\overline{Y}_{2\bullet} - \overline{Y}_{\bullet\bullet})^2 \\ &= r_1\left(\overline{Y}_{1\bullet} - \frac{r_1\overline{Y}_{1\bullet} + r_2\overline{Y}_{2\bullet}}{r_1 + r_2}\right)^2 + r_2\left(\overline{Y}_{2\bullet} - \frac{r_1\overline{Y}_{1\bullet} + r_2\overline{Y}_{2\bullet}}{r_1 + r_2}\right)^2 \\ &= \frac{r_1 r_2^2}{(r_1 + r_2)^2}(\overline{Y}_{1\bullet} - \overline{Y}_{2\bullet})^2 + \frac{r_2 r_1^2}{(r_1 + r_2)^2}(\overline{Y}_{1\bullet} - \overline{Y}_{2\bullet})^2 \\ &= \frac{r_1 r_2}{(r_1 + r_2)}(\overline{Y}_{1\bullet} - \overline{Y}_{2\bullet})^2 \\ &= \frac{1}{1/r_1 + 1/r_2}(\overline{Y}_{1\bullet} - \overline{Y}_{2\bullet})^2 \end{aligned}$$

The pooled variance is related to the within-group residual sum-of-squares via

$$S^2_{\mathrm{pooled}} = \frac{\mathrm{SS}_{\mathrm{resid}}}{r_1 + r_2 - 2}$$

Hence, the F-ratio given by

$$F = \frac{\mathrm{SS}_{\mathrm{group}}/1}{\mathrm{SS}_{\mathrm{resid}}/(r_1 + r_2 - 2)} = \frac{(\overline{Y}_{1\bullet} - \overline{Y}_{2\bullet})^2}{S^2_{\mathrm{pooled}}(1/r_1 + 1/r_2)} = T^2$$

where $T = \frac{\overline{Y}_{1\bullet} - \overline{Y}_{2\bullet}}{S_{\mathrm{pooled}}\sqrt{1/r_1 + 1/r_2}}$ is the usual two-sample t-statistic.

(d) Under the null hypothesis $H_0 : \mu_1 = \mu_2$, the T statistic has a $t_{r_1+r_2-2}$ distribution and the F statistic has an $\mathcal{F}_{1,r_1+r_2-2}$ distribution.

(e) We see that an $\mathcal{F}_{1,d}$ distribution with numerator degree-of-freedom 1 and denominator degree-of-freedom d is exactly identical to the square of a t_d distribution with d degrees-of-freedom.

(f) A two-sample t-test can test one-sided alternative hypotheses such as $H_1 : \mu_1 > \mu_2$ or $H_1 : \mu_1 < \mu_2$. The ANOVA can only test the two-sided alternative $H_1 : \mu_1 \neq \mu_2$.

Exercise 4 (Working with different ANOVA parametrisations)

In class, we introduced the one-way ANOVA model using the **mean parametrization** as

$$Y_{ji} = \mu_j + \epsilon_{ji} \ , \quad j = 1, 2, ..., J, \ i = 1, 2, \ldots, r_j \ ,$$

where $\epsilon_{ji} \overset{\text{iid}}{\sim} N(0, \sigma^2)$ across both indices j and i. Under this parametrization, the model parameters μ_j are interpreted as the population group means of each group j.

Let's consider two other equivalent parametrizations of the *same* model. For simplicity, we consider the balanced case with $r_1 = r_2 = \ldots = r_J = r$ for all groups.

The "sum constraint" parametrization:

$$Y_{ji} = \mu + \alpha_j + \epsilon_{ji} \ , \quad j = 1, 2, ..., J, \ i = 1, 2, \ldots, r \ , \text{ with } \textstyle\sum_{j=1}^{J} \alpha_j = 0 \ .$$

(a) Under the sum constraint parametrization, μ is interpreted as the overall mean of all groups (i.e., $\mu = (\mu_1 + \mu_2 + \ldots + \mu_J)/J$ and each α_j is interpreted as the difference between the mean of group j and the overall mean (i.e., $\alpha_j = \mu_j - \mu$ for each j).

(b) The joint likelihood function is given by

$$L(\mu, \alpha_1, \alpha_2, \ldots, \alpha_J, \sigma^2 | \boldsymbol{Y}) = \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^{Jr} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{j=1}^{J} \sum_{i=1}^{r} (Y_{ji} - \mu - \alpha_j)^2 \right\} \ ,$$

where Jr is the total sample size across all J groups. Maximising the likelihood L is equivalent to maximising the log-likelihood,

$$l = \log L = -(Jr/2) \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{j=1}^{J} \sum_{i=1}^{r} (Y_{ji} - \mu - \alpha_j)^2 + \text{constant} \ .$$

Differentiating the log-likelihood with respect to μ and setting to 0 gives

$$0 = \sum_{j=1}^{J} \sum_{i=1}^{r} (Y_{ji} - \mu - \alpha_j) \qquad \Longleftrightarrow \qquad Jr\,\mu = \sum_{j=1}^{J} \sum_{i=1}^{r} Y_{ji} - r \sum_{j=1}^{J} \alpha_j \ .$$

Because we are using the sum constraint, $\sum_{j=1}^{J} \alpha_j = 0$, the MLE for μ is therefore given by

$$\hat{\mu} \ = \ \frac{1}{Jr} \sum_{j=1}^{J} \sum_{i=1}^{r} Y_{ji} \ = \ \overline{Y}_{\bullet\bullet} \ .$$

Similarly, differentiating the log-likelihood with respect to α_j and setting to 0 gvies

$$0 = \sum_{i=1}^{r} (Y_{ji} - \mu - \alpha_j) \qquad \Longleftrightarrow \qquad r\,\alpha_j = \sum_{i=1}^{r} Y_{ji} - r\,\mu \ .$$

Hence, the MLE for α_j is given by

$$\hat{\alpha}_j \ = \ \frac{1}{r} \sum_{i=1}^{r} Y_{ji} - \hat{\mu} \ = \ \overline{Y}_{j\bullet} - \overline{Y}_{\bullet\bullet} \ .$$

Finally, differentiating the log-likelihood with respect to σ^2 and setting to 0 gives

$$0 = -\frac{Jr}{2\sigma^2} + \frac{1}{2(\sigma^2)^2}\sum_{j=1}^{J}\sum_{i=1}^{r}(Y_{ji}-\mu-\alpha_j)^2 \qquad \Longleftrightarrow \qquad \sigma^2 = \frac{1}{Jr}\sum_{j=1}^{J}\sum_{i=1}^{r}(Y_{ji}-\mu-\alpha_j)^2 \ .$$

Hence, the MLE for σ^2 is given by

$$\hat{\sigma}^2 = \frac{1}{Jr}\sum_{j=1}^{J}\sum_{i=1}^{r}(Y_{ji}-\hat{\mu}-\hat{\alpha}_j)^2 = \frac{1}{Jr}\sum_{j=1}^{J}\sum_{i=1}^{r}(Y_{ji}-\overline{Y}_{j\bullet})^2$$

(c) The fitted means $\hat{\mu}+\hat{\alpha}_j$ for each group j under the sum parametrization are given by

$$\hat{\mu}+\hat{\alpha}_j \ = \ \overline{Y}_{\bullet\bullet} + \overline{Y}_{j\bullet} - \overline{Y}_{\bullet\bullet} \ = \ \overline{Y}_{j\bullet} \ ,$$

precisely the group sample means.

The "contrast" parametrization:

$$Y_{ji} = \mu + \alpha_j + \epsilon_{ji} \ , \quad j = 1, 2, ..., J, \ i = 1, 2, \ldots, r \ , \ \text{with } \alpha_1 \equiv 0 \ .$$

(d) Under the contrast parametrization, μ represents the mean of group 1 (i.e., $\mu \equiv \mu_1$) and each α_j represents the difference between the mean of group j and group 1 (i.e., $\alpha_j = \mu_j - \mu_1$ for $j = 2, 3, \ldots, J$).

(e) Using analogous calculations to part (b) above, we find that the MLEs of the model parameters under the contrast parametrization are

$$\begin{aligned} \hat{\mu} &= \overline{Y}_{1\bullet} \ , \\ \hat{\alpha}_j &= \overline{Y}_{j\bullet} - \overline{Y}_{1\bullet} \ , \quad j = 2, \ldots, J \ , \\ \hat{\sigma}^2 &= \frac{1}{Jr}\sum_{j=1}^{J}\sum_{i=1}^{r}\left(Y_{ji} - \overline{Y}_{j\bullet}\right)^2 \ . \end{aligned}$$

(f) Under the contrast parametrization, the fitted means $\hat{\mu}+\hat{\alpha}_j$ for each group j are given by

$$\hat{\mu}+\hat{\alpha}_j \ = \ \overline{Y}_{1\bullet} + \overline{Y}_{j\bullet} - \overline{Y}_{1\bullet} \ = \ \overline{Y}_{j\bullet} \ ,$$

precisely the group sample means as before.

By comparing your results from parts (c) and (f), and also the expressions for $\hat{\sigma}^2$ under both parametrizations, convince yourself that these two parametrizations are fitting the *exact same* model to the data.